MIL-V-16721D
20 August 1980
SUPERSEDING
MIL-V-16721C
19 February 1971

MILITARY SPECIFICATION

# VALVES, FOOT, LOW PRESSURE, CAST-IRON (FOR WATER)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE AND CLASSIFICATION

1.1 Scope This specification covers threaded-end and flanged-end cast-iron foot valves for use with low-pressure water.

1.2 Classification. The valve shall be of one type, and of the following nominal pipe sizes as required or specified (see 6.2):

1-1/2-inch
2-inch
3-inch
4-inch
6-inch

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-V-3 - Valves, Fittings, and Flanges (Except for Systems Indicated Herein); Packaging of.

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables For Inspection By Attributes.

(Copies of specification, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4820

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A126 - Gray Iron Castings for Valves, Flanges, and Pipe Fittings.

(Application for copies should be addressed to the American Sosiety For Testing And Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B16.1 - Cast Iron Pipe Flanges and Flanged Fittings Class 125.

(Application for copies should be addresssed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

SP-25 - Standard Marking System For Valves, Fittings, Flanges and Unions.

(Application for copies should be addressed to the Manufacturers Standardization Society of the Valve and Fittings Industry, 1815 North Fort Myer Drive, Arlington, Va. 22209.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Description. The foot valve shall consist of an iron body complete with a removable strainer bolted to the inlet, a single leather-faced or rubber-faced disk or flap, or a bronze disk, and a seat. The complete valve assembly shall be suitable for operation at steady pressures not exceeding 25 pounds per square inch (psi).

3.2 First article. When specified (see 4.2.1 and 6.3), the contractor shall furnish a sample valve for first article inspection. This inspection shall be performed by the contractor and shall consist of the examination and tests specified in 4.2.1. The first article inspection shall be scheduled and administered in accordance with the applicable provisions of the contract.

3.3 Interchangeability. All foot valves of the same classification, furnished with similar options under any specific contract, shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, and spare parts. No deviations shall be acceptable without prior written approval of the contracting officer.



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3.4 Standard commercial product. The foot valve of the same classification shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the unit being furnished. A standard commercial product is a product which, has been sold or is currently being offered for sale, on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.5 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.5.1 Gray iron castings. Cast iron shall be in accordance with the requirements of ASTM A126, class A (ordinary gray cast iron), shall be of uniform quality, free from blowholes, porosity, hard spots, shrinkage defects, cracks, or other injurious defects. Repairs shall not be made on the castings.

3.6 Design. Foot valves shall be designed to permit flow in one direction only, and shall be fabricated for ease of accessiblity for maintainance and servicing in the field. When specified (see 6.2), flanged foot valves shall be in accordance with ANSI B16.1.

3.7 Construction. The foot valve shall be constructed in accordance with the requirements specified herein.

3.7.1 Body. The body shall be cast iron designed to withstand a working pressure of 25 psi and internal test pressure of 125 psi.

3.7.2 Seat. The seat shall be bronze or cast iron. The flow area at the seat shall be approximately equal to the inside area of a pipe of the nominal valve size.

3.7.3 Disk. The disk or flap shall be bronze or cast-iron-faced with leather or with natural or synthetic rubber. The leather or rubber shall have a minimum thickness of 3/32 inch, and may be used as the disk hinge. The disk shall be so mounted as to open and close without binding, and seat properly. The leather or rubber shall be mounted for ready removal and replacement.

3.7.4 Strainer. The strainer shall be of cast iron, or a weldment, and shall be bolted to the body. The open area of the strainer shall not be less than 1-1/4 times the flow area of a pipe of the nominal valve size. The open slot width shall be a maximum of 7/16 inch.

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3.8 Performance. The foot valve when used as specified herein, shall perform without undue shock or jar. The valve shall permit flow in one direction only and shall close tightly to prevent flow in the opposite direction. The motion of operating components shall be smooth and nonsticking.

3.9 Treatment and painting. Unless otherwise specified (see 6.2), the valves shall be treated and painted in accordance with the manufacturer's standard practice.

3.10 Spare parts. When specified (see 6.2), spare parts shall be provided.

3.11 Identification marking. Unless otherwise specified (see 6.2), valves shall be marked in accordance with MSS SP-25.

3.12 Workmanship. Workmanship shall be such as to produce foot valves that are in accordance with the requirements of this specification and insure proper functioning of all parts of the unit.

3.12.1 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided. All bolts, nuts, and screws shall be tight.

3.12.2 Machine work. Tolerances shall be such as to insure proper functioning of fits for the assembled valve. Finished surfaces, as applicable, shall be level, square, and concentric.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) First article inspection (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).
(c) Packaging inspection (see 4.7).

4.2.1 First article inspection. First article inspection shall be performed on one valve when first article sample is required (see 3.2, 6.2, and 6.3). This inspection shall include the examination of 4.5 and the tests of 4.6. The first article may be a standard production item from the contractor's current inventory provided the valve meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining valves to be furnished under the contract.

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4.2.2 Quality conformance inspection Quality conformance inspection shall be performed on the sample valves selected in accordance with 4.4. This inspection shall include the examination of 4.5 and the tests of 4.6.

4.3 Inspection lot. All valves of the same style and size offered for delivery at one time shall be considered one lot for the purpose of inspection. The sample unit shall be one complete valve.

4.4 Sampling. A random sampling of valves shall be selected from each lot in accordance with MIL-STD-105.

4.4.1 Sampling for examination. Examination of the valves shall be based on inspection level II, and an Acceptable Quality Level (AQL), of 2.5 percent defective for major defects and 4.0 percent for minor defects.

4.4.2 Sampling for tests. Tests of the valves shall be based on inspection level S-3 and an AQL of 4.0 percent defective.

4.5 Examination. The first article, when furnished, and each sample selected in accordance with 4.4.1 shall be examined for compliance with the requirements of Section 3 of this specification. The examination shall be as specified in Table I.

TABLE I. Classification of defects.

| Categories | | References |
|---|---|---|
| Critical: | | |
| Major: | | |
| 101 | Components missing, not as specified. | (see 3.1) |
| 102 | Interchangeability not maintained. | (see 3.3) |
| 103 | Material not as specified, damaged or defective. | (see 3.5 and 3.5.1) |
| 104 | Construction not as specified. | (see 3.7) |
| Minor: | | |
| 201 | Treatment and painting not as specified. | (see 3.9) |
| 202 | Identification and marking missing incorrect or illegible, or not as specified. | (see 3.11 thru 3.11.3) |

4.6 Tests. The first article, when furnished, and each sample selected in accordance with 4.4.2 shall be tested to determine compliance with this specification. Tests shall be conducted as specified herein.

4.6.1 Performance test. The valves shall be tested under simulated operating conditions to determine that the equipment conforms to the requirements of this specification and is satisfactory for use. Failure to meet the requirements or to perform satisfactorily shall be cause for rejection.

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4.6.2 Hydrostatic test. The valves shall be tested hydrostatically for seat tightness. The internal chamber of the valve body shall be filled with water and a pressure of 125 psi shall be applied. Leakage shall be cause for rejection. Weepage through the disk face is not considered leakage.

4.7 Packaging inspection. The preservation-packaging, packing, and marking of the valves shall be inspected to verify conformance to the requirements of Section 5.

5 PACKAGING

5.1 Preservation-packaging, packing, and marking. Unless otherwise specified (see 6.2), the valve shall be preserved, packaged, packed and marked in accordance with MIL-V-3. The level of preservation-packaging and level of packing shall be as specified (see 6.2).

6 NOTES

6.1 Intended use. Foot valves covered by this specification are intended for suction lines connected to self-priming centrifucal pumps. The valves are recommended for low-pressure cold water use where the pressure does not exceed 34-foot head (approximately 15 pounds).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2).
(c) When a first article sample is required (see 3.2, 4.2.1, and 6.3).
(d) When flanged foot valves are required (see 3.6).
(e) Treatment and painting, if different (see 3.9).
(f) Spare parts required (see 3.10).
(g) When marking other than MSS SP-25 is required (see 3.11).
(h) Level of preservation-packaging and level of packing required (see 5.1).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the Defense Acquisition Regulations (DAR). The first article should be a first production item consisting of one complete foot valve unit or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The contracting officer should include specific instructions in all acquisition instruments, regarding arrangement for examinations, tests, and approval of the first article.

6.4 Material requirements. Records of chemical analyses and tension tests on materials furnished in accordance with ASTM specifications are systematically maintained by the material producer. However, materials for valve components are usually produced well in advance of orders and are furnished to the valve industry from the contractor's current stocks. Valve manufacturers maintain a general quality certification of the materials used without specific reference to the lot and samples representative of the particular

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stock being used. If the quantity and criticality of valves furnished under this specification warrant production of special lots of materials which can be identified with specific tests for chemical and tensile requirements, the contract or order should state that records of these specific tests shall be made available to the procuring agency by the valve manufacturer.

6.5 Contract data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulation (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - ME
Navy - YD
Air Force - 99

Preparing activity:

Navy - YD

Project No. 4820-0317

Review activity:

DLA - CS

User activities:

Army - CE
Navy - MC, OS

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